

# PIYUSH VISHWAKARMA

Cybersecurity Enthusiast | SOC Analyst Aspirant | Digital Forensics Learner

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## PROFESSIONAL SUMMARY

Cybersecurity-focused MCA student specializing in Information Security, with hands-on experience in threat detection, network packet analysis, log monitoring, file integrity monitoring (FIM), and digital forensics. AWS Certified Cloud Practitioner with practical skills in Python-based security tooling, Linux administration, and SOC concepts. Developed 5 security-focused Python projects demonstrating real-world capabilities in network reconnaissance, host-based intrusion detection, and security event analysis. Actively targeting Cybersecurity Intern, SOC Analyst Intern, and Information Security Analyst roles.

## TECHNICAL SKILLS

|             |                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security    | Threat Detection & Monitoring, Log Analysis & SIEM Concepts, File Integrity Monitoring (FIM), Packet Analysis, Vulnerability Assessment, Network Reconnaissance, Incident Response Fundamentals |
| Networking  | TCP/IP, DNS, SSH, HTTP/S, OSI Model, Port Scanning, Scapy, Wireshark (Basic)                                                                                                                    |
| Programming | Python (Scapy, Socket Programming, SHA256 Hashing, Log Parsing), Bash/Shell Scripting, SQL                                                                                                      |
| Cloud & OS  | AWS (EC2, S3, IAM, VPC, Security Groups, CloudWatch), Linux (CLI, systemctl, chmod/chown, User Mgmt), Windows                                                                                   |
| Tools       | Git, GitHub, VS Code, Docker (Basic), MongoDB                                                                                                                                                   |
| Forensics   | Digital Forensics Concepts, Log Parsing & Analysis, Evidence Collection Fundamentals                                                                                                            |

## CYBERSECURITY PROJECTS

### Network Packet Sniffer

*Python | Scapy | Network Analysis*

GitHub: <https://github.com/ItsPiyushVishwakarma/Python-Network-Packet-Sniffer>

- Built a live network traffic capture tool using Python and Scapy, capturing source/destination IPs, protocols (TCP, UDP, ICMP), packet sizes, and timestamps — simulating core SOC network monitoring workflows.
- Reinforced understanding of packet analysis, OSI model layers, and traffic pattern recognition used in real-world threat detection and network forensics.

### File Integrity Monitoring (FIM) Tool

*Python | SHA256 | Endpoint Security*

GitHub: [github.com/ItsPiyushVishwakarma/Python-File-Integrity-Checker](https://github.com/ItsPiyushVishwakarma/Python-File-Integrity-Checker)

- Developed a Python-based FIM solution using SHA256 cryptographic hashing to detect unauthorized file modifications — directly replicating Host-Based Intrusion Detection System (HIDS) functionality.
- Automated baseline hash storage with timestamps and generated integrity verification reports, aligning with endpoint security monitoring and compliance auditing practices.

### Python Port Scanner

*Python | Socket Programming | Recon*

GitHub: [github.com/ItsPiyushVishwakarma/Python-Port-Scanner](https://github.com/ItsPiyushVishwakarma/Python-Port-Scanner)

- Engineered a TCP port scanner using Python socket programming to identify open ports and enumerate active network services on target hosts.
- Applied network reconnaissance methodology to understand attack surface mapping, service identification, and vulnerability discovery fundamentals.

### Python Log Analyzer

*Python | Log Parsing | SIEM Concepts*

GitHub: [github.com/ItsPiyushVishwakarma/Python-Log-Analyzer](https://github.com/ItsPiyushVishwakarma/Python-Log-Analyzer)

- Created a log analysis tool to parse system/application logs, detect suspicious events, and categorize entries (errors, warnings, critical) — directly applicable to Level 1 SOC analyst alert triage and SIEM workflows.
- Automated detection of anomalous patterns simulating security event correlation used in Security Operations Centers.

**Password Strength Checker**

Python | Authentication Security

GitHub: [github.com/ItsPiyushVishwakarma/Password-Strength-Checker](https://github.com/ItsPiyushVishwakarma/Password-Strength-Checker)

- Built a password complexity evaluator assessing uppercase, lowercase, digits, special characters, and length with a weighted strength score and visual meter — demonstrating authentication security and access control concepts.

**CERTIFICATIONS**

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| <b>AWS Certified Cloud Practitioner</b> – Amazon Web Services   Score: 731   Apr 2025<br><b>Career Essentials in System Administration</b> – Microsoft & LinkedIn   Mar 2025<br><b>Career Essentials in Generative AI</b> – Microsoft & LinkedIn   Mar 2025<br><b>Generative AI &amp; Thoughtful Online Search</b> – LinkedIn   Jan 2025 |  |
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**EDUCATION**

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| <b>Master of Computer Applications (MCA) – Cybersecurity</b><br><i>Chhatrapati Shivaji Maharaj University, Panvel, Maharashtra   Pursuing</i> <ul style="list-style-type: none"> <li>                             Focus areas: Information Security, Network Security, Digital Forensics, Cryptography, Ethical Hacking.                         </li> </ul> | 2025 – 2027 |
| <b>Bachelor of Science – Information Technology (BSc-IT)</b><br><i>Tilak College, Mumbai University, Mumbai, Maharashtra   Completed</i>                                                                                                                                                                                                                     | 2021 – 2024 |

**LEADERSHIP & ACTIVITIES**

|                                                                                                                                                                                                                                                                                                                                                                                                               |             |
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| <b>Technical Head – Council Member</b><br><i>Tilak College, Navi Mumbai</i> <ul style="list-style-type: none"> <li>                             Led technical operations for 10+ large-scale college events, coordinating cross-functional teams of 500+ students across audio-visual and digital infrastructure — building incident coordination and operational skills.                         </li> </ul> | 2021 – 2024 |
| <b>NSS (National Service Scheme) Candidate</b><br><i>Tilak College, Navi Mumbai</i> <ul style="list-style-type: none"> <li>                             Secured 2nd Place in the Institutional Presentation Competition (Academic Year 2022–2023).                         </li> </ul>                                                                                                                        | 2022 – 2024 |

Open to Work: Cybersecurity Intern | SOC Analyst Intern | Information Security Intern | Junior Security Analyst